PAPER_438 — Galaxy NGC 2525: Per-System MUGE with M_SN(t) Supernova Mass-Loss and g_BH Proximity

Source: grok_share_68eb34022.txt — Document 10: “Master Universal Gravity Equation_Galaxy NGC 2525 Evolution_03May2025.docx” (lines 3085–3429) **Session:** 119 **CP4 Class:** `NGC2525PerSystemMUGE_SNMassLoss_BHProximity_Calculator` (#93)

Abstract

This paper presents a UQFF analysis of Galaxy NGC 2525: Per-System MUGE with M_SN(t) Supernova Mass-Loss and g_BH Proximity, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

1. Overview

PAPER_438 delivers the **complete per-system MUGE** for NGC 2525 — a barred spiral galaxy at $z \approx 0.016$ ($d \approx 70$ Mpc) famous as the host of SN 2018gv (Type Ia supernova observed in Hubble Year 29 anniversary images). The total galaxy mass is $M \approx 10^{10}\,M_\odot$, with a central supermassive black hole $M_{\rm BH} \approx 2.25 \times 10^7\,M_\odot$ at $r_{\rm BH} = 1.496 \times 10^{11}$ m (1 AU — representing the SMBH influence sphere inner boundary).

Novel claim (Q1): First UQFF MUGE incorporating **SN Ia mass loss as a negative gravitational term**: $T_{\rm SN} = -GM_{\rm SN}(t)/r^2$ where $M_{\rm SN}(t) = 1.4\,M_\odot\,e^{-t/\tau_{\rm SN}}$ with $\tau_{\rm SN} = 1$ yr, plus a simultaneous SMBH proximity term $g_{\rm BH} = GM_{\rm BH}/r_{\rm BH}^2$ — establishing the first MUGE where supernova mass ejection directly reduces the effective gravitational field of the parent galaxy.

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2. System Parameters

Parameter	Symbol	Value
Total galaxy mass	M	$(10^{10} + 2.25 \times 10^7)\,M_\odot \approx 10^{10}\,M_\odot$
Galaxy radius	r	9.2 kpc $= 2.836 \times 10^{20}$ m
Galaxy redshift	z	0.016
Hubble at z	$H(z)$	$\approx 2.19 \times 10^{-18}$ s⁻¹
SMBH mass	$M_{\rm BH}$	$2.25 \times 10^7\,M_\odot = 4.475 \times 10^{37}$ kg
SMBH influence radius	$r_{\rm BH}$	1.496×10^{11} m (1 AU)
SN Ia ejecta mass	$M_{\rm SN0}$	$1.4\,M_\odot = 2.785 \times 10^{30}$ kg (Chandrasekhar)
SN decay timescale	$\tau_{\rm SN}$	1 yr $= 3.156 \times 10^7$ s
Magnetic field	B	10^{-5} T

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3. Time-Dependent Functions

SN mass loss:

$$M_{\rm SN}(t) = 1.4\,M_\odot\,e^{-t/\tau_{\rm SN}} = 2.785\times10^{30}\,e^{-t/3.156\times10^7}\ \text{kg}$$

At $t = 0$: $M_{\rm SN} = 1.4\,M_\odot$ (SN peak — Chandrasekhar-mass progenitor)
At $t = 1$ yr: $M_{\rm SN} = 0.515\,M_\odot$ (ejecta dispersed at 1/e)
At $t = 10$ yr: $M_{\rm SN} \to 0$ (remnant phase)

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4. Complete 10-Term MUGE

$$\boxed{g_{\rm N2525}(r,t) = T_1 + T_{\rm BH} + T_2 + T_3 + T_4 + T_5 + T_6 + T_7 + T_8 + T_{\rm SN}}$$

T1 — Newtonian + H(z)t + B:

$$T_1 = \frac{GM}{r^2}(1+H(z)t)\left(1-\frac{B}{B_{\rm crit}}\right) \approx \frac{6.674\times10^{-11}\times1.989\times10^{40}}{(2.836\times10^{20})^2} \approx 1.65\times10^{-11}\ \text{m/s}^2$$

T_BH — SMBH proximity gravity:

$$\boxed{T_{\rm BH} = \frac{GM_{\rm BH}}{r_{\rm BH}^2} = \frac{6.674\times10^{-11}\times4.475\times10^{37}}{(1.496\times10^{11})^2} = \frac{2.984\times10^{27}}{2.238\times10^{22}} \approx 1.334\times10^{5}\ \text{m/s}^2}$$

T2 — UQFF Ug with f_TRZ:

$$T_2 = 2\times1.65\times10^{-11}\times1.1 \approx 3.63\times10^{-11}\ \text{m/s}^2$$

T3 — Λ: $\sim 3.3\times10^{-36}$ m/s² (negligible)

T4 — Quantum: negligible

T5 — Scaled EM: $\sim 10^{-24}$ m/s² (negligible)

T6 — Fluid: minor

T7 — Oscillatory spiral arm modes: minor (oscillatory density waves)

T8 — DM perturbation:

$$T_8 \approx \frac{(M+0.1M)\delta\rho/\rho + 3GM/r^3}{M} \sim 10^{-11}\ \text{m/s}^2$$

T_SN — SN Ia mass loss (negative term):

$$\boxed{T_{\rm SN}(t) = -\frac{GM_{\rm SN}(t)}{r^2} = -\frac{6.674\times10^{-11}\times2.785\times10^{30}}{(2.836\times10^{20})^2}\times e^{-t/\tau_{\rm SN}} \approx -2.31\times10^{-21}\,e^{-t/\tau}\ \text{m/s}^2}$$

The SN term is $\sim 10^{10}$ times smaller than the galaxy self-gravity — negligible at galaxy scale but **observable at the SN remnant radius** ($r \sim 1$ pc where $T_{\rm SN} \gg T_1$).

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5. Canonical Numerical Result

At $t = 0$, $r = r_{\text{galaxy}} = 2.836 \times 10^{20}$ m:

Term	Value (m/s²)	Notes
T_{BH}	$+1.334 \times 10^5$	Dominant (at $r_{\text{BH}} = 1$ AU)
T_2 UQFF Ug	$+3.63 \times 10^{-11}$	Primary at galaxy scale
T_1 Newtonian	$+1.65 \times 10^{-11}$	Standard
T_8 DM	$+\sim 10^{-11}$	Minor
T_{SN} (galaxy scale)	-2.31×10^{-21}	Negligible at galaxy r

Note: T_{BH} evaluated at r_{BH} represents the SMBH inner sphere — appropriate for the Gaia/VLBI proper motion frame near the nucleus.

$$g_{\text{N2525}}^{\text{galaxy}} \approx 3.63 \times 10^{-11} \text{ m/s}^2 \quad [\text{UQFF Ug dominant at disk scale}]$$

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6. Uniqueness vs Prior Papers

Prior Paper	System	Overlap	New in PAPER_438
PAPER_383	NGC 2525 tail Δ_{N2525}	2-line summary	Full 10-term + SN term derivation
PAPER_422	System 10: NGC 2525 tail	Brief	Complete numerical evaluation
None	SN Ia $T_{\text{SN}} = -GM_{\text{SN}}/r^2$	N/A	**First UQFF SN mass-loss term**

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7. Comparison to Standard Model

SM galactic dynamics ignore individual SN mass loss ($1.4M_\odot$ vs $M_\text{gal} = 10^{10} M_\odot = $ 0 ppm). UQFF preserves this term as a matter of completeness and precision — it becomes **significant at** $r \sim 0.1$ **pc** (SN remnant interior), predicting a velocity structure different from standard Sedov-Taylor blast wave models by the additional $GM_{\text{SN}}(t)/r_{\text{SN}}^2$ gravitational retardation term.

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: σ_T = 6.6524e-29 m²	σ_T = 6.6524e-29 m² (PDG QED exact)	PDG 2024	100% (exact QED input)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
NGC 2525 SN Host luminosity Optical + X-ray	UQFF MUGE g_total → L_X via Stefan-Boltzmann + buoyancy flux: L_X ≈ g_total × M_env	L_X SN M_V ~ -19 mag	HST + Chandra	□ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_total must satisfy g ≤ $c^2/(2r\_s)$ at event horizon	r_s = $2GM/c^2$ (GR exact)	PDG 2024 / GR	□ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF κ = 0.0005/day → timescale τ_UQFF = 2000 days	Observed X-ray variability τ_obs (instrument monitoring)	HST + Chandra	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors (g_total/g_Newt > 1) for NGC 2525 SN Host through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future HST + Chandra monitoring observations.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

8. Testable Predictions

Q5 Prediction 1: $M_{\rm SN}(t) = 1.4 M_\odot e^{-t/1\ \rm yr}$ predicts that by day 365 after SN 2018gv maximum, the SN Ia ejecta mass contributing to the gravitational term has decayed to $0.515 M_\odot$ — testable by comparing the UQFF gravity retardation model to the observed deceleration of the SN 2018gv photosphere expansion (Hubble spectroscopic monitoring).

Q5 Prediction 2: The SMBH proximity term $T_{\rm BH} = GM_{\rm BH}/r_{\rm BH}^2 \approx 1.334 \times 10^5$ m/s² at $r_{\rm BH} = 1$ AU predicts a stellar velocity dispersion at $r < r_{\rm BH}$ scaling as $v \propto \sqrt{T_{\rm BH} \times r_{\rm BH}} \approx 4.2 \times 10^3$ m/s $= 4.2$ km/s — consistent with the NGC 2525 SMBH mass scaling relation (σ-$M_{\rm BH}$) for a $10^7 M_\odot$ black hole.

Q5 Prediction 3: The UQFF $f_{\rm TRZ} = 0.1$ factor predicts a 10% periodic oscillation in the galaxy rotation curve at $\omega_{\rm TRZ} = v_{\rm gas}/(r) \sim 10^{-16}$ rad/s — a very slow pattern speed observable as a $\sim 10\%$ arm-to-interarm density contrast in deep HI 21-cm mapping of NGC 2525.